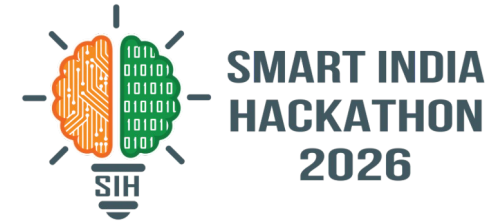
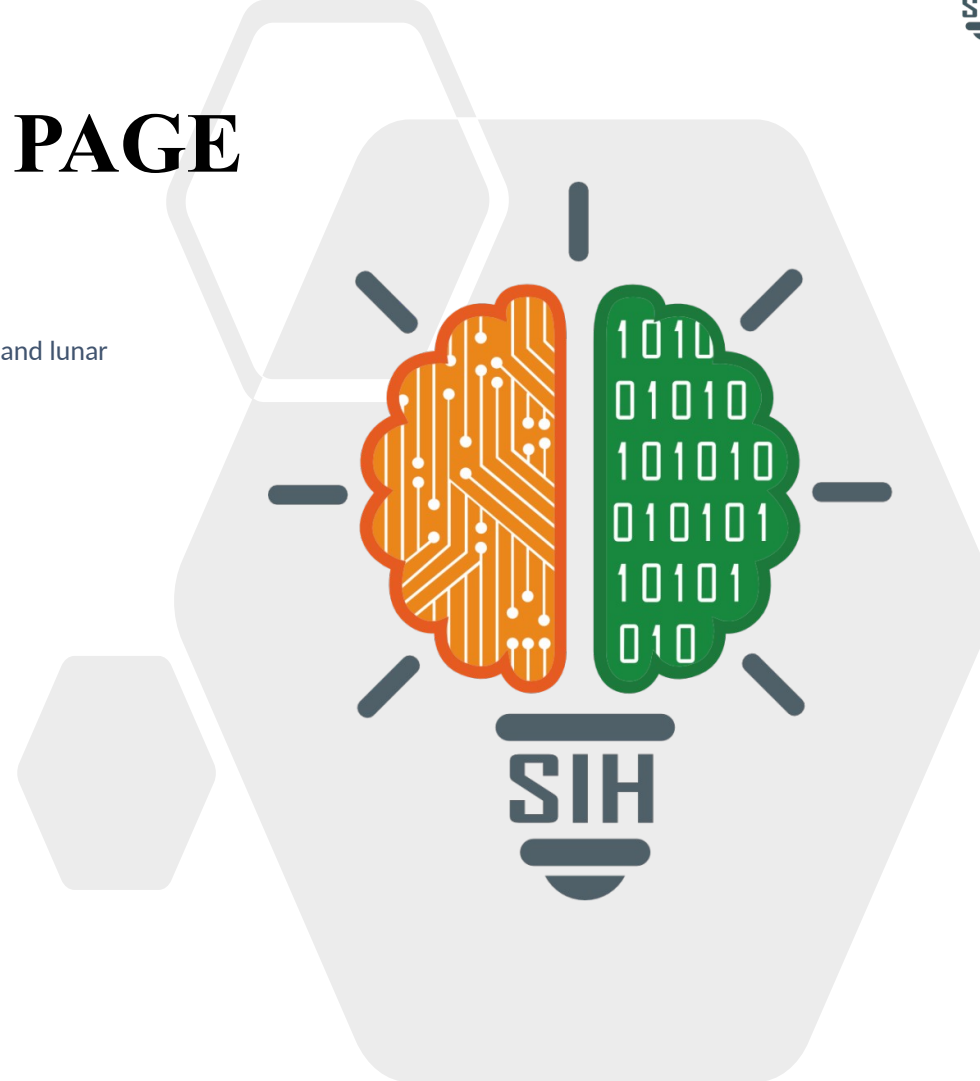


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** - 26166
- **Problem Statement Title** - Sub-pixel correspondence between Chandrayaan-2 optical images and lunar reference images, robust to illumination, viewpoint and scale
- **Theme** - Space Technology
- **PS Category** - Software
- **Team ID** - 800C4B
- **Team Name** - Bugs Janta Party



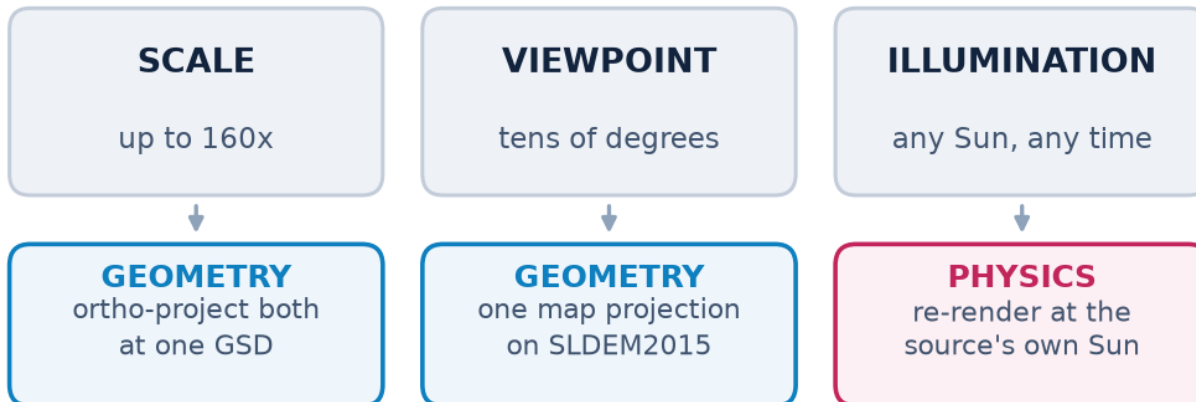
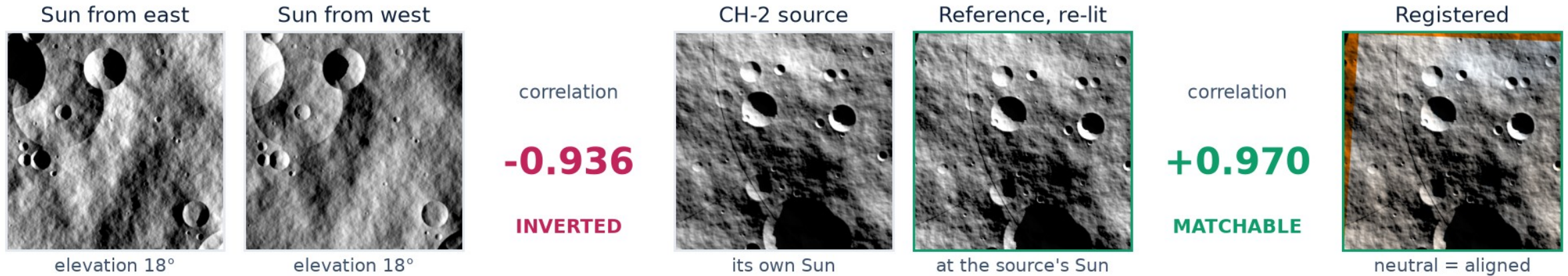
SETU

Geometry for scale and viewpoint. Physics for the Sun. Learning only for what is left.

SETU: spend the geometry, remove the Sun

THE PROBLEM

WHAT SETU DOES



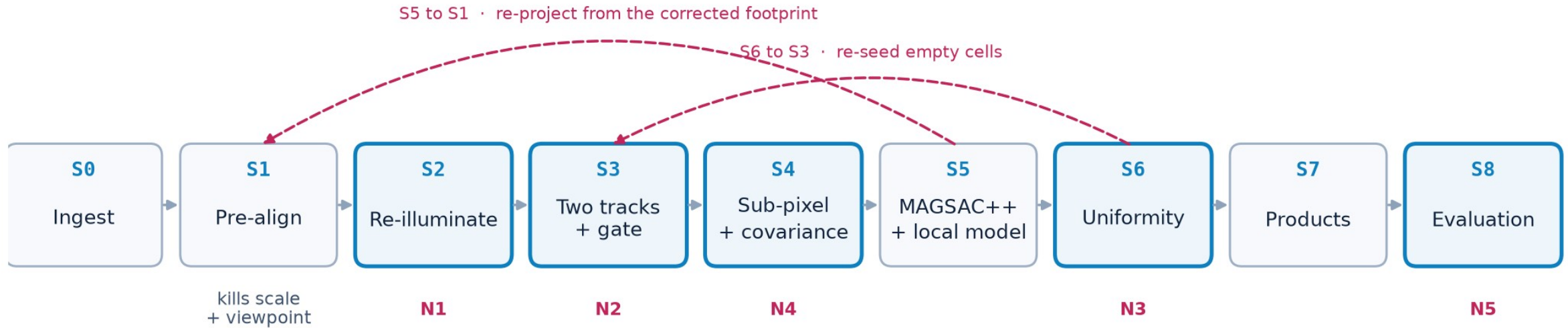
MEASURED ON 31 PAIRS WITH EXACT GROUND TRUTH

0.361 px	100%	416×	0.81
tie-point RMSE vs truth	matches within 3 px	better than SIFT	coverage at 150 pts

Every match carries a 2×2 covariance. It weights the fit, sets the outlier threshold, and ships in the tie-point file.

Uniformity is an objective, not a by-product. Lattice quota over the true overlap; empty cells re-seeded.

TECHNICAL APPROACH



Raster / geodesy

NumPy SciPy rasterio GDAL pyproj

Planetary I/O

pds4_tools pvl spiceypy

Classical CV

OpenCV MAGSAC++ scikit-image

Structural

phase congruency MIM CFOG LNIFT

Deep matching

PyTorch kornia LoFTR LightGlue XFeat

Serve / ship

FastAPI React Vite Docker CI

THREE CHOICES THAT MATTER

- **Never upsample the coarser image.** Work at the coarser GSD, reach it by area-averaging.
- **Parallax is not optional off-nadir.** At 25° and 200 m of relief it is 373 OHRC pixels.
- **Harness built before the matcher.** No tuning decision was made against intuition.

FEASIBILITY AND VIABILITY

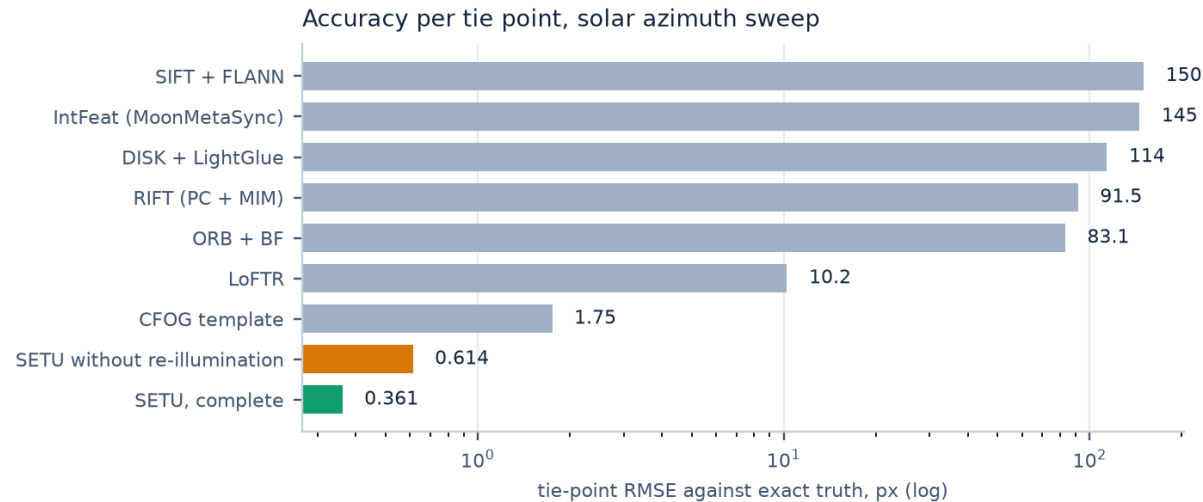
AT AN 8× SCALE RATIO, SETU DECLINES RATHER THAN LIES

144**0.0%****0****no registration returned**

confident matches from the deep matcher of them correct · median error 92 px

accepted by the agreement gate

refusing to answer beats a plausible wrong answer



RISK

PRADAN needs an account,
archive is 200 GB

SPICE / ISIS integration
can eat days

59 m DEM cannot render
25 cm OHRC photorealistically

Repetitive craters make deep
matchers confidently wrong

MITIGATION

Built against the controlled
benchmark; readers written and tested

Tier B corner fit is the default;
Tier A degrades and says so

Render constrains mid-frequency;
sub-pixel comes from the real pair

The agreement gate, measured
at 0.0% precision rejected

BUILT, NOT DESCRIBED

9 / 9

stages, both feedback edges

41

tests, green

5.6 s

per pair, no GPU

1

Docker image, CI builds it

12

methods benchmarked

IMPACT AND BENEFITS

0.361 px

tie-point RMSE vs exact truth

100%

matches within 3 px of truth

100%

inlier ratio after MAGSAC++

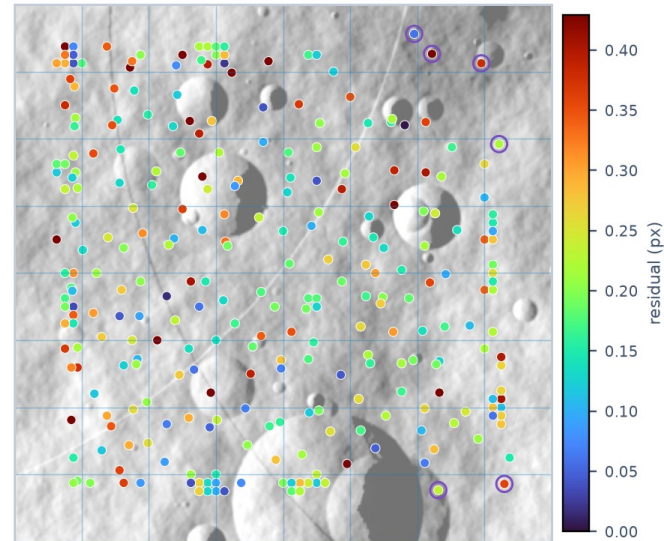
0.81

coverage, mean over 31 pairs

416×

better per point than SIFT

283 tie points over the 8×8 lattice



coverage 0.95 on this run · purple ring = re-seeded cell

WHAT THIS BUYS ISRO

- **An archive that co-registers.** Kilometre-level geolocation is why OHRC, TMC-2 and IIRS are hard to combine.
- **A tie-point list that can be ingested.** Per-point covariance, so a bundle adjustment can weight it.
- **Pointing stability, for free.** The per-row jitter spline falls out of registration.
- **Generalises past the Moon.** Any airless body with a shape model has the same problem.



TRY IT LIVE

<https://arya-patil686.github.io/setu-sih2026/>

Three registrations at increasing difficulty, the pipeline played stage by stage, tie points coloured by residual, and the full leaderboard.

RESEARCH AND REFERENCES

RESEARCH THIS BUILDS ON

- **He et al.** MatchAnything, TPAMI 2026, arXiv:2501.07556. Track A weights.
- **Li, Hu, Ai.** RIFT, IEEE TIP 29:3296, 2020. Track B descriptor.
- **Ye et al.** HOPC, IEEE TGRS 55(5) 2017; CFOG, ISPRS 2019.
- **Kovesi.** Image Features from Phase Congruency, 1999. Vendored.
- **Kumar et al.** MoonMetaSync, arXiv:2410.11118, 2024. The lunar floor we beat.
- **Tungathurthi.** arXiv:2602.14993, 2026. Documents the 4 to 6 km OHRC error.
- **Barker et al.** SLDEM2015, Icarus 273:346, 2016. The shape model.
- **Chowdhury et al.** OHRC in-orbit performance, Current Science 118(4):560, 2020.
- **Verma et al.** IIRS thermal emission correction, Icarus 383:115075, 2022.
- **Barath et al.** MAGSAC++, CVPR 2020. With an adaptive threshold from our covariances.

DATA

Chandrayaan-2 L1 via PRADAN · LRO NAC and NAC DTMs via ODE · SLDEM2015 at 512 ppd · Kaguya TC via JAXA DARTS · LROC WAC



LIVE DEMO

<https://arya-patil686.github.io/setu-sih2026/>

<https://github.com/Arya-Patil686/setu-sih2026>

REPRODUCE EVERY NUMBER

[python experiments/run_sweeps.py](#)

[python scripts/build_demo_bundle.py](#)

Ground truth is exact by construction: both images of every pair are rendered from one terrain model under a known transform. Results are on that benchmark; the PDS4 and PDS3 readers and the SPICE path await archive access, not code.